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Week 5

Mathematics Pedagogy / Physics Pedagogy
MTH3004M

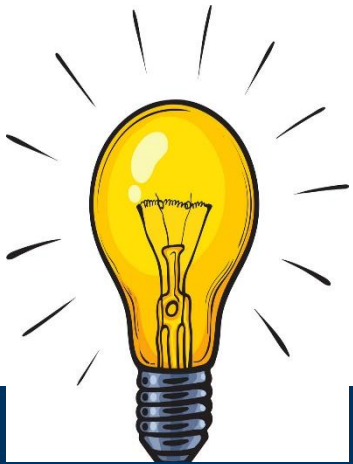


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From Effective Teaching and Learning Theories to Misconceptions

Thinking about Experimentation
and the other *Theories of*
Learning these gave rise to

Behaviourism



Watch and take notes on what stands out to you in this video

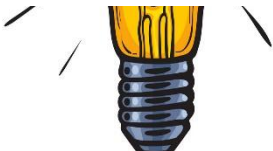


SLT Video

Watch and take notes on what stands out to you in this video



<https://www.youtube.com/watch?v=zerCK0IRjp8>



Strength and Weaknesses of



<u>Strengths</u>	<u>Weaknesses</u>
This is a classic study	Ethics
A controlled experiment that is reliable, as it can be replicated	The situation was artificial and the children were aware they were part of a study. This could have led to demand characteristics.
Has many important real world implications	Weapons effect in the case of presenting the children with the mallet
	Young children know the difference between play or fantasy violence and the real thing

SLT Video



- 1) The children were put in a strange situation, exposed to some unusual adult behaviour and given toys to play with which encouraged them to act unnaturally.
- 2) Cumberbatch (1990) found that children who had not played with a Bobo Doll before were five times as likely to imitate the aggressive behaviour than those who were familiar with it; he claims that the novelty value of the doll makes it more likely that children will imitate the behaviour.



Social Learning Theory

Observational learning is a result of witnessing another person doing something.

The learning is performed later and cannot be explained as having been taught in any other way.

This type of learning also encompasses the concept of behaviour avoidance as a result of seeing another person behave in a certain way and receive a negative consequence.



The four processes

The four processes of observational learning are:

Process 1: Attention

- Before students can imitate a model's behaviour, they must pay attention to what the model is doing or saying.

Process 2: Retention

- To produce a model's action, students must be able to store the model's action in their memory for future retrieval. Students' retention will be improved when a teacher gives vivid, logical, and clear demonstrations.

Process 4: Production

- To attending and remembering, students must be physically capable of reproducing the model's action. Here, the students need a lot of practice, feedback, and coaching before they can reproduce the model's action.

Process 4: Motivation

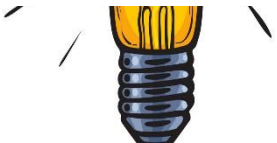
- The students must be motivated to demonstrate the model's action. Reinforcement can be used to encourage observational learning. This may also be indirect where a student may see other students being reinforced for a particular behaviour and then increase their own production of that behaviour.



Reciprocal

Or ***Reciprocal Determinism***

The world and an individual's behaviour *cause each other*. Behaviourism essentially states that it is the environment causing a specific behaviour. According to Bandura (1977) though, this is too simplistic. Behaviour causes environment as well as environment causes behaviour.



Reciprocal



Behaviour

Individual's actions,
choices and verbal
statement



Personal factors

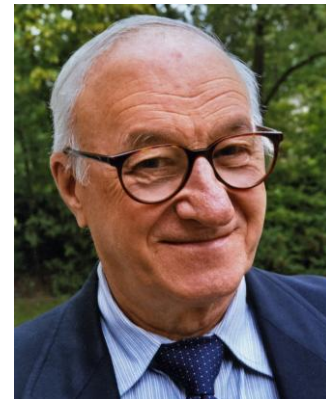
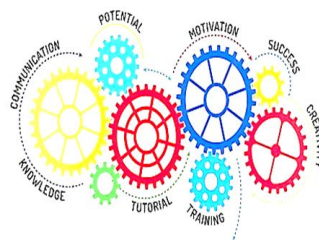
Person's expectations,
beliefs, attitudes,
strategic thinking, and
intelligence



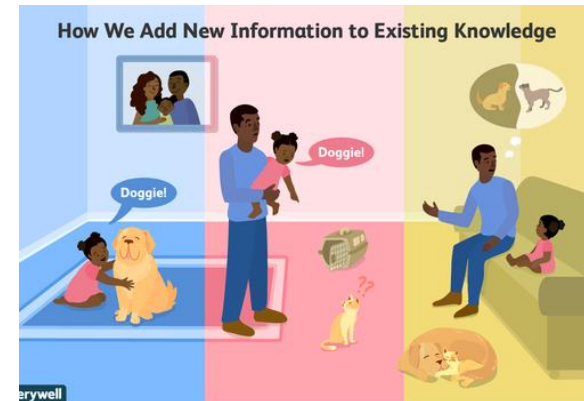
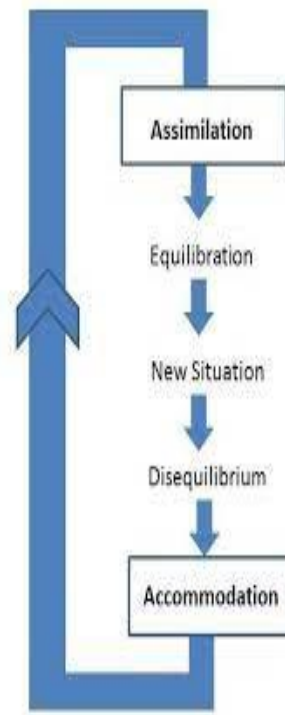
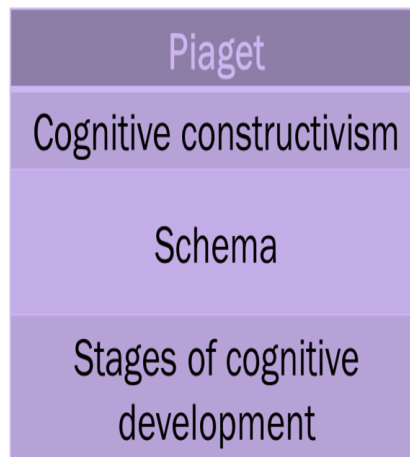
Environmental factors

Resources,
consequences of
actions, other people,
and physical setting

Lecture Quick



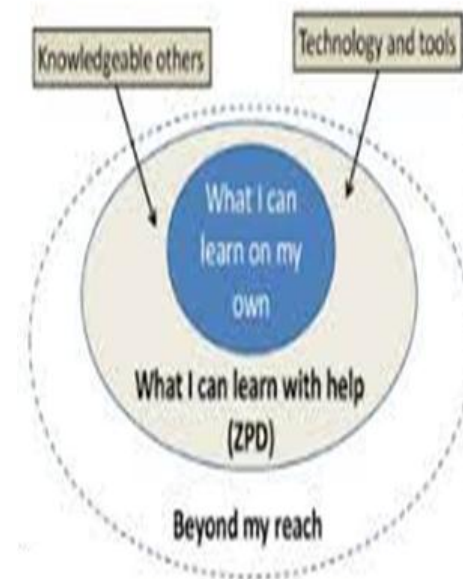




Piaget
Cognitive constructivism
Schema
Stages of cognitive development

Vygotsky
Social constructivism
Zone of proximal development- MKOs
Scaffolding

ZPD and scaffolding

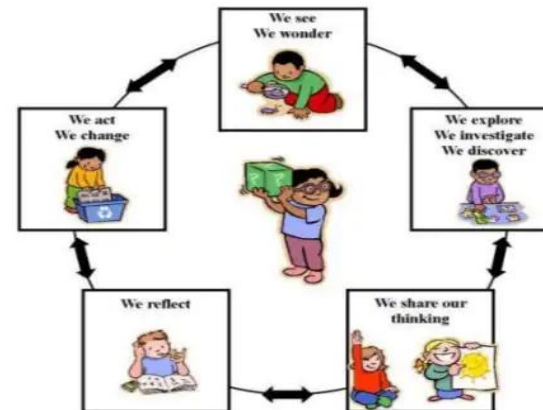


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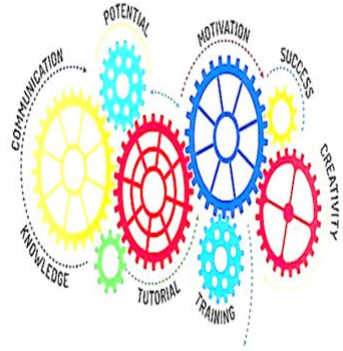
Bandura
Observation
Imitation
Modelling

Bruner

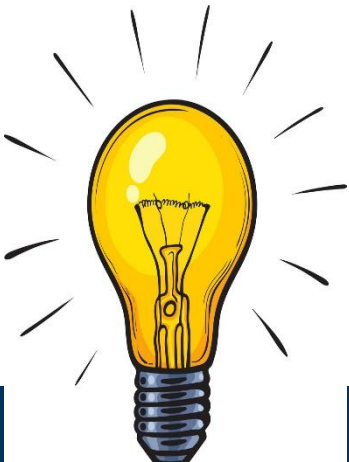


Activity on Learning theories

What would you do differently?



In pairs/groups, use any of the learning theories as your guide to consider how you might improve students' learning on a topic they may have difficulties in.



Activity on Learning theories

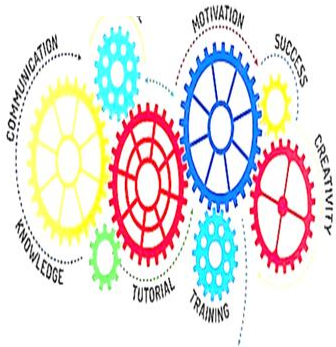
What would you do differently?

Some key features to apply:

- Modelling
- 4 processes (attention, retention, production, motivation)
- That 'people are social creatures' and learn through some form of human interaction
- Cognitive Development
- Behaviourism and Conditioning

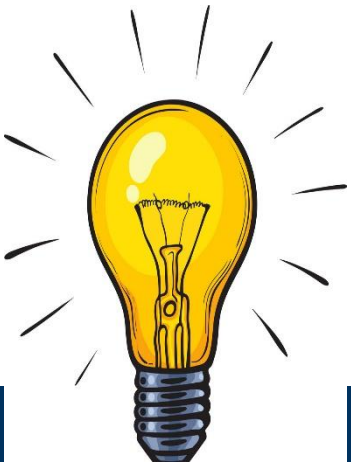
Some possible strategies:

- promoting role models within the school or community;
- undertaking and observing roleplay;
- school trips to see something;
- incorporate social media and educational technology deliberately;
- mentoring schemes;
- devise a way to practically demonstrate that subject content;
- exchange programmes



Activity on learning Theories

Be creative and very specific about both the problem in learning you are hoping to overcome and your strategy to improve this.

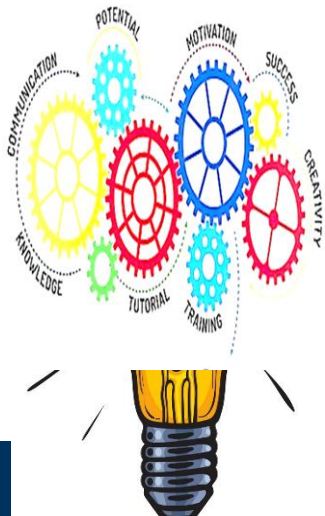


Activity on learning Theories

Post your answers of what
could have been done
differently (Maths experience)



<https://padlet.com/nfotou/activity-on-learning-theories-zetvrlr4i6mwhy0m>

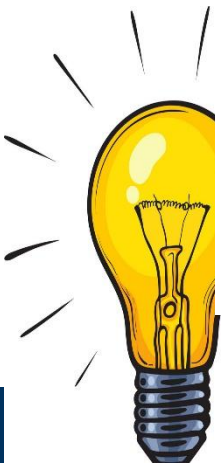


Activity on learning theories

**An example of
a learning experience**



School students usually struggle with basic algebra in mathematics (Gersten et al., 2009). How would that be related to SLT and/or constructivism?



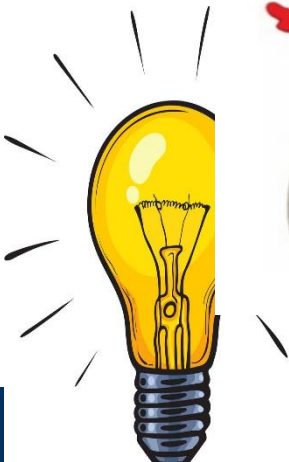
Activity on learning theories

**An example of
a learning experience**



Unmotivated teacher?

We were taught practice sheets with lots of repetition but there was no quick progress made. Teachers would usually explain it a few times to the whole class, using the white board...



Activity on learning Theories

An example of what could have been done differently (Maths experience)

- Maths teacher being a good role model by showing how passionate she/he is and trying to spread the passion to the students.
- Make use of group work activities where a more able student who is motivated and responsible acts as a role model for students who do not care about maths or they are not good at maths.





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Misconceptions in Science and Mathematics

Effective Teaching (Pedagogy)

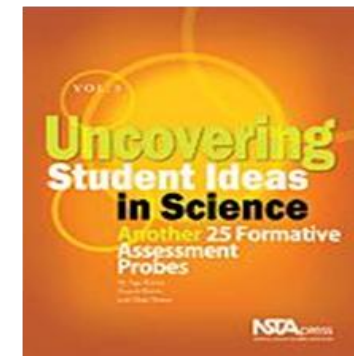
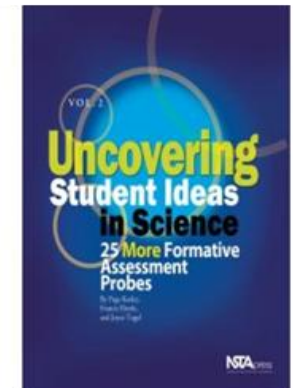
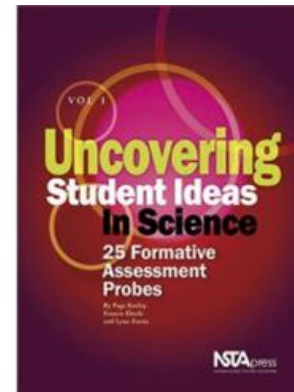
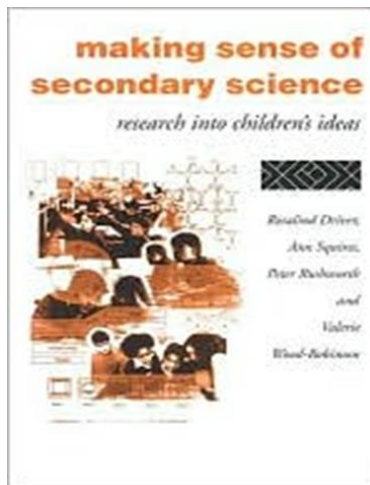
Below is a list of claims from Husbands and Pearce (2012):

1. Consideration of pupil voice
2. Depends on teachers' behaviour, knowledge and understanding, and beliefs
3. Clear thinking of longer-term learning outcomes and short-term goals
4. Build on pupils' prior learning and experience
5. Involve scaffolding pupil learning
6. Involve a range of techniques
7. Develop higher order thinking and metacognition - good use of dialogue and questioning in order to do so
8. Embed assessment for learning
9. Are inclusive and take the diverse needs of a range of learners, as well as matters

Try and answer the following two questions:

1. How would you define a misconception?
2. What impact can it have on students' learning?

Students' Misconceptions and how to Address them: An introduction



Just because you
taught it doesn't mean
they learned it

Misconceptions

- ▶ A misconception is an idea or view inconsistent with the scientific understanding because it is based on faulty thinking or understanding.

Alternative
conceptions

Intuitive ideas

Preconceptions

Erroneous beliefs

Preconceived
notions

Non-scientific
concepts

Experience is NOT always
the best teacher.

Misconceptions are tenacious—when
they make sense to students, they're
VERY hard to change.

Key characteristics of Misconceptions

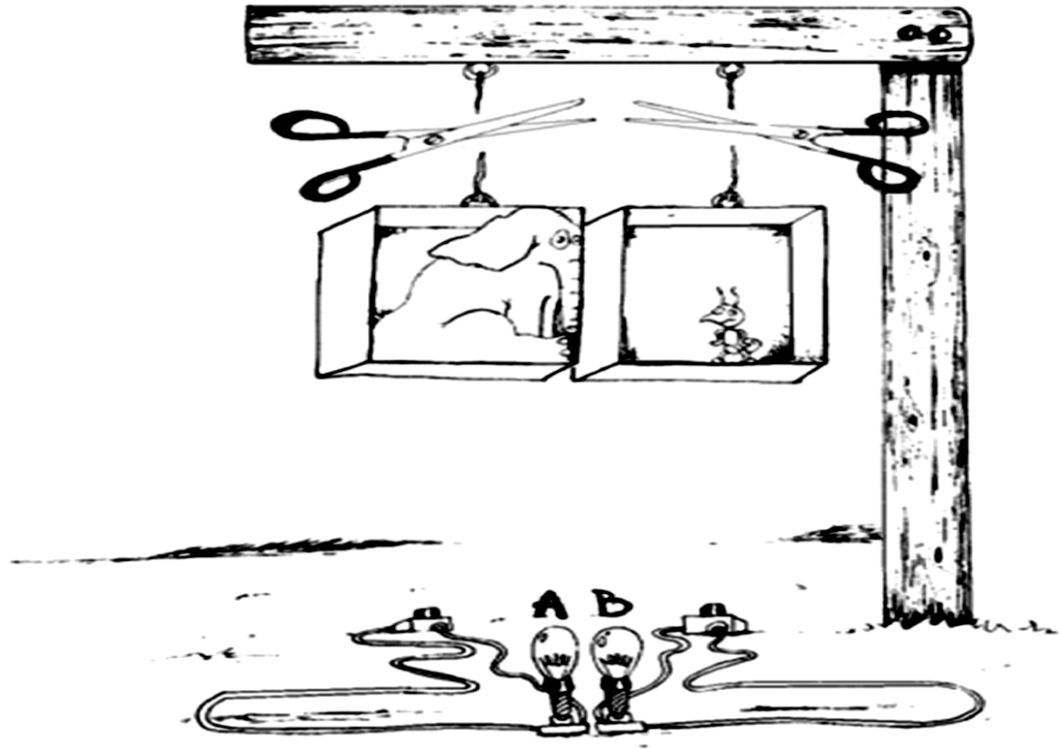
- misconceptions  students' ideas being inconsistent with the scientific understanding.
- Students' misconceptions:
 1. are strongly held
 2. differ from experts' notions
 3. have a strong influence in students' understanding

Types of Misconceptions

- Misconceptions are ‘popular’ conceptions rooted in everyday observations (*Preconceived notions*).
- Misconceptions include views learned by students from sources other than scientific education, such as religious or mythical teachings (*Non-scientific beliefs*).
- Misconceptions also arise when students are taught scientific information in a way that does not help them to deal with their own preconceived notions and non-scientific beliefs or when they overgeneralise information (*Conceptual misunderstandings*).
- Misconceptions arise from the use of words that mean one thing in everyday life and another in a scientific context (*Vernacular misconceptions*).
- Misconceptions can be falsities often learned at an early age and retained unchallenged into adulthood (*Factual misconceptions*).

Examples:

Preconceived notions



If the ropes shown in the figure are cut at the same time, will the bulbs be switched on at the same time or will one of them be first?

A) Both at the same time

B) Bulb A first

C) Bulb B first

Task 1

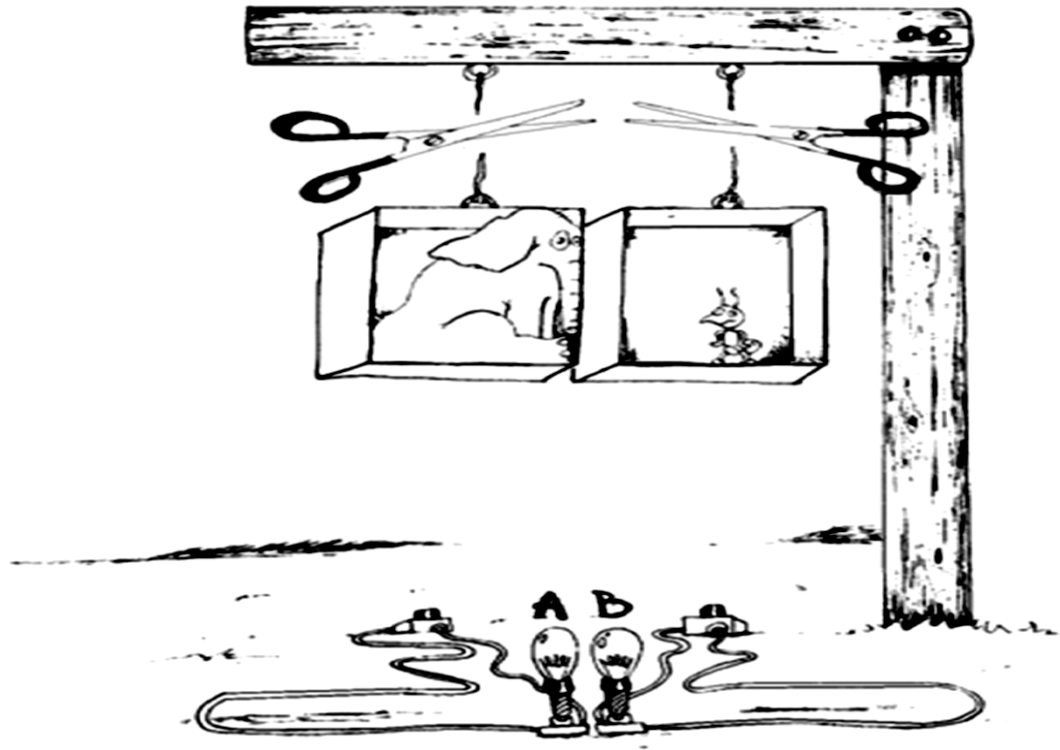
Think-Try to think of this situation and make a prediction

Pair- Try to explain this to the person next to you

Share- Share your prediction and explanation with the rest of the class

Examples:

Preconceived notions



If the ropes shown in the figure are cut at the same time, will the bulbs be switched on at the same time or will one of them be first?

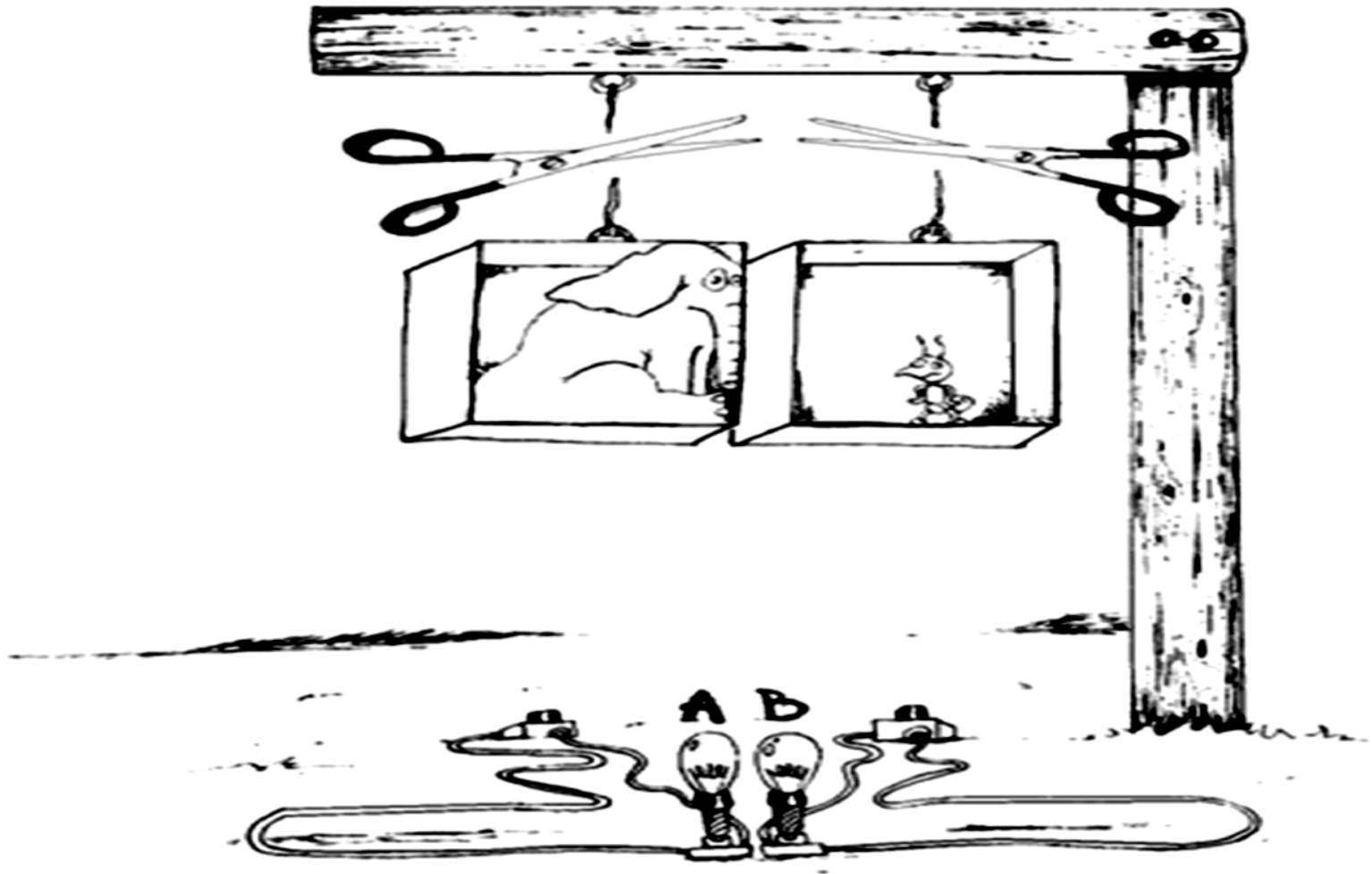
A) Both at the same time

B) Bulb A first

C) Bulb B first

Task 2

- Provide a simple explanation to a first year student who is not aware of and can't use any formulas / equations.



If the ropes shown in the figure are cut at the same time, will the bulbs be switched on at the same time or will one of them be first?

A) Both at the same time

B) Bulb A first

C) Bulb B first



NO Air Resistance

- An object will accelerate if the forces acting upon it are unbalanced.
- The acceleration of an object is directly related to the net force and inversely related to its mass.
- Applied to the elephant-mouse scenario, the box with the elephant experiences a much greater force (which tends to produce larger acceleration).
- Yet, the mass of an object resists acceleration.
- Thus, the greater mass of the elephant (which tends to produce smaller acceleration) offsets the influence of the greater force. It is the force/mass ratio which determines the acceleration.

Air resistance

- The amount of acceleration is directly proportional to the amount of net force (unbalanced forces) acting upon it. Falling objects initially gain speed because there is no force ***big enough*** to balance the downward force of gravity.
- Yet as an object gains speed, it encounters an increasing amount of upward air resistance force. In fact, objects will continue to accelerate (**gain speed**) until the air resistance force increases to a ***large enough value to balance the downward force of gravity***.
- Since the box with the elephant has more mass, it weighs more and experiences a greater downward force of gravity. The box with elephant will have to accelerate (gain speed) for a longer period of time before there is sufficient upward air resistance to balance the large downward force of gravity.

Do heavy objects fall faster than light objects?



https://www.youtube.com/watch?v=jme_vSj5wRo

1. Newton's 1st law:

An object will accelerate if the forces acting upon it are unbalanced.

1. Newton's 2nd law :

The amount of acceleration is directly proportional to the amount of net force (unbalanced force) acting upon it.

1. Terminal velocity:

The terminal velocity is the final and highest velocity attainable occurring when the sum of the drag force (F_d) and the buoyancy is equal to the downward force of gravity (F_g) acting on an object falling in a fluid.

The box with elephant falls faster than the feather because it never reaches a terminal velocity. It continues accelerating accumulating more and more air resistance, approaching a terminal velocity without actually reaching it.

The box with the ant reaches a terminal velocity in an early stage of its fall as there is not much air resistance required to cease its acceleration. Once that terminal velocity reached, the remainder of this box fall will occur with that small terminal velocity.

Non-scientific beliefs

https://www.youtube.com/watch?v=K9vIK2_I22U

Conceptual misunderstandings

Misconceptions in mathematics usually originate in prior instruction as students incorrectly generalize prior knowledge

Apply whole-number knowledge to compare numbers

3.471 and 3.7



$3.471 > 3.7$

Incorrectly apply general knowledge of fractions (focusing on the size of the smallest decimal place in the numbers but **ignoring the relative value** of the digits in those locations) to compare numbers

3.471 and 3.2



$3.471 < 3.2$

Vernacular misconceptions

The concept of Energy

- Things use up energy
- We/ I am are running out of Energy (conservation?).

Factual misconceptions

Houseflies live for only 24 hours



The common housefly (or *Musca domestica*) tends to live for about two to three weeks, but can live up to a couple months. And if you consider the animal's entire life cycle, the housefly lives for days even before sprouting wings.

Lightning ***never*** strikes the same place twice



Lightning can actually strike anywhere twice - the main question is the probability of this occurring.

Task 3

In pairs discuss

Why do we have seasons?

Task 3 continued

- In your pairs, think about why we have seasons



<https://www.youtube.com/watch?v=p0wk4qG2mlg>



<https://www.youtube.com/watch?v=b3TRUDKpoAs>

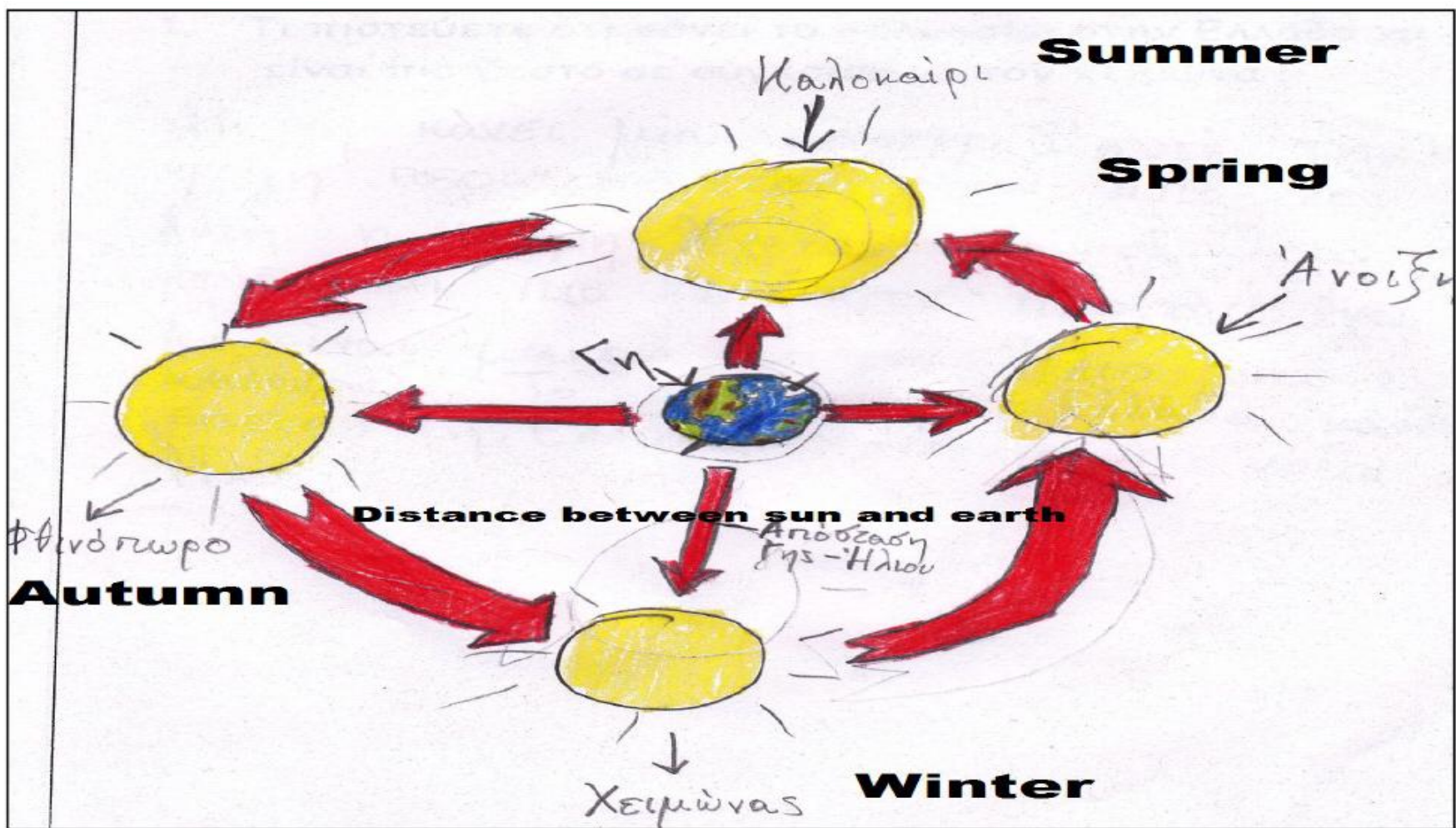


Figure 1. A student's typical drawing explaining: (a) seasons by showing the Sun moving closer to the Earth in summer and further away in winter (Fotou, 2012, p.28)

Examples

- ***Preconceived notions***

Seasons are caused by the distance between the Sun and Earth

- ***Non-scientific beliefs***

There is a dark side of the moon

Conceptual misunderstandings

Blood in your veins is blue

- ***Vernacular misconceptions***

Mass and weight

- ***Factual misconceptions***

Houseflies live for only 24 hours

<https://www.youtube.com/watch?v=LqaDf2fuUH8>

Yesterday's research-based findings on students' misconceptions

1. Students come in science and mathematics classrooms with a diverse set of ideas concerning natural objects, events and phenomena. Each subfield within the different disciplines seems to have its set of ideas.
2. These set of ideas students bring in the classroom cut across age, gender, ability and sociocultural boundaries.
3. These ideas are tenacious and show resistant to change by traditional teaching strategies.
4. The ideas that students hold often parallel explanations and theories offered by previous generations of scientists and philosophers.

5. The ideas students bring in classroom have their roots in a diverse set of personal experiences including direct observations and their interpretations, peer culture-language and teachers' instruction.
6. Teachers often hold the same ideas as their students.
7. These ideas students bring in classroom interact with knowledge presented in formal instruction resulting in a diverse variety of unintended learning outcomes.
8. There has been a variety of instruction approaches suggested as classroom tools that can facilitate **conceptual change.**

How to help students get over their Misconceptions

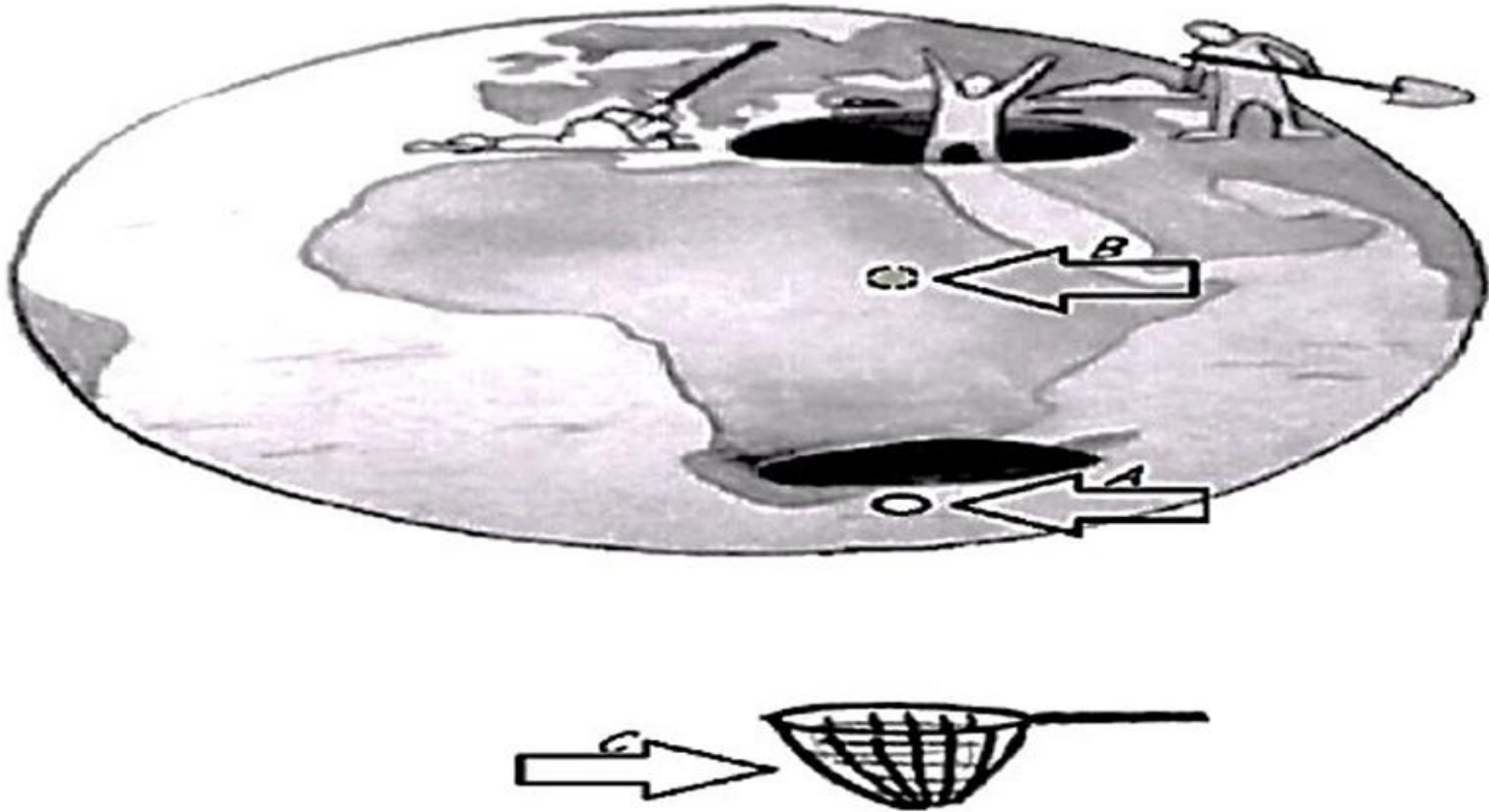
1. Assessing misconceptions

When presenting new information to students, it is helpful to assess any misconception they may hold on the material.

For example, when teaching students about climate change:

- Understandings of the distinction between weather and climate.
- Knowledge about the concept of "deep time."

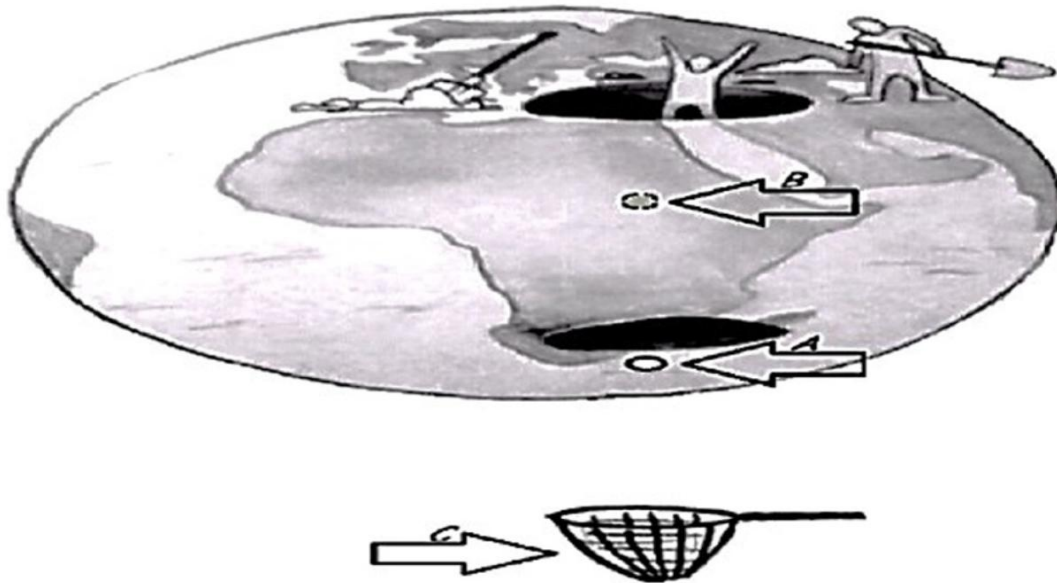
How to do the assessment



Two people have dug a huge tunnel straight down from the one side of the Earth to the other (as shown in the figure) and one of them jumps in. Where will this person stop?

A) On the other side of the tunnel B) In the middle of the tunnel
C) In the net

- Pollev.com/nikolaosfoto472



Two people have dug a huge tunnel straight down from the one side of the Earth to the other (as shown in the figure) and one of them jumps in. Where will this person stop?

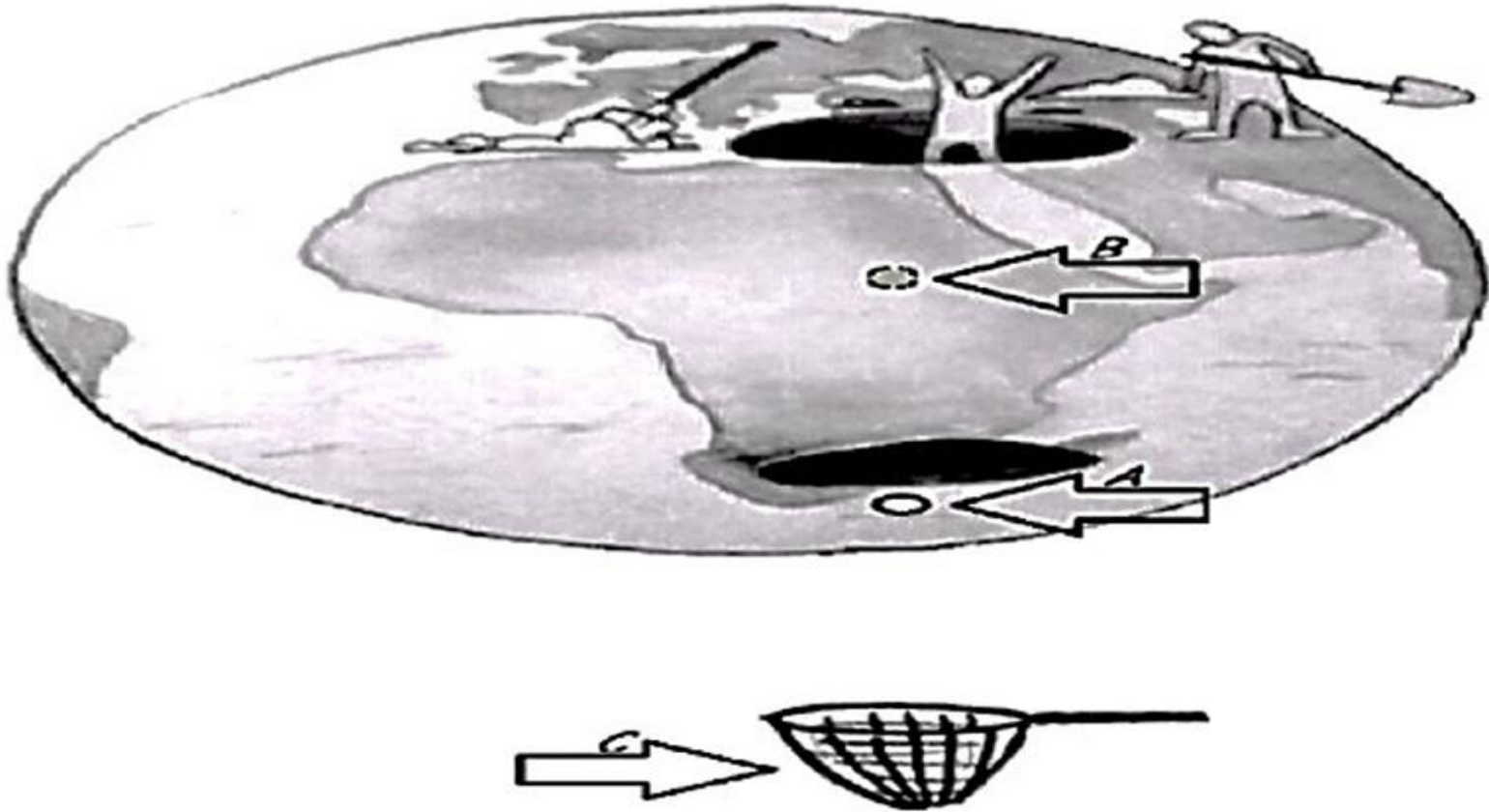
A) On the other side of the tunnel

B) In the middle of the tunnel

C) In the net

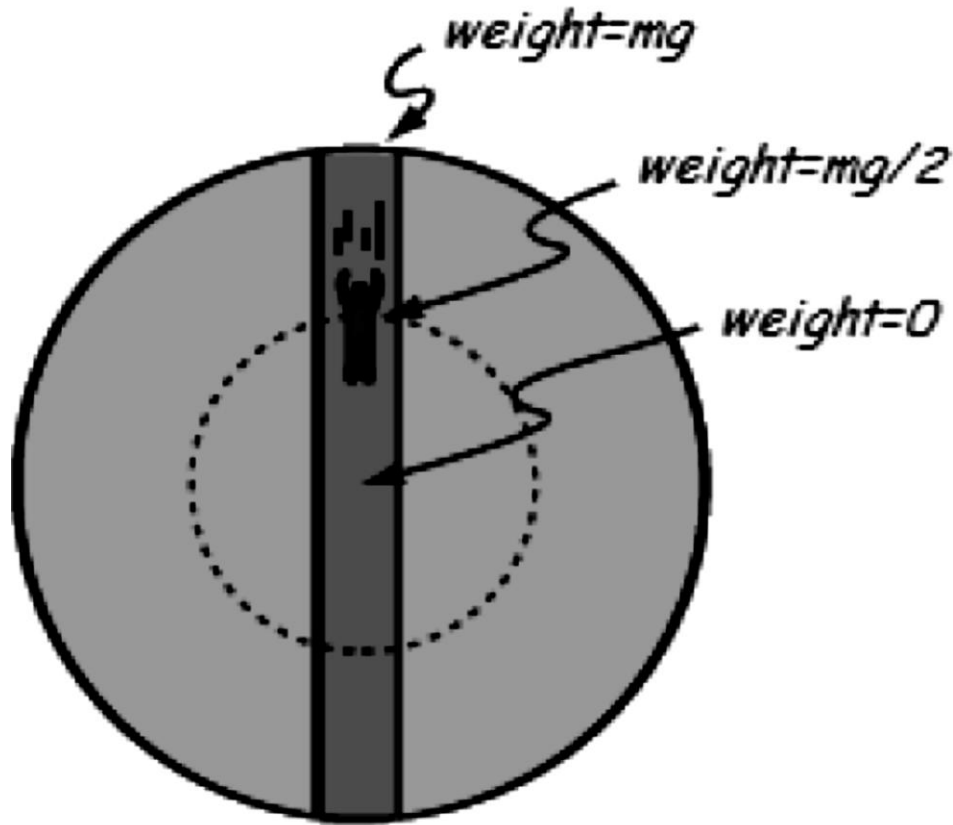
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How to do the assessment



Two people have dug a huge tunnel straight down from the one side of the Earth to the other (as shown in the figure) and one of them jumps in. Where will this person stop?

- A) On the other side of the tunnel
- B) In the middle of the tunnel
- C) In the net



The initial acceleration of the person who jumped into the tunnel would be the acceleration of gravity on the surface of the Earth.

$$g = \frac{GM_{\text{Earth}}}{R_{\text{Earth}}^2} = 9.8 \text{ m / s}^2$$

This becomes progressively smaller as the person approaches the centre of the Earth.

Choose a topic of interest and design a diagnostic assessment for any misconceptions student may hold as per this topic.

2. Building on misconceptions

Tapping onto Prior Knowledge

Misconceptions are not only a barrier to learning, when used properly they could serve a productive purpose in the classroom.

It is important to consider which components of their already acquired knowledge could be beneficial in building a more robust understanding of new, scientific concepts.

Misconceptions are not bad! They can be a good thing when teachers **PURPOSEFULLY** use them to bridge students' ideas with conceptual understanding of science.



Use what students know and think to help them get to the other side

Ordinary instructional approaches are not effective in altering science and math content misconceptions

Pfundt & Duit (1991).

Conceptual Change

In presenting new concepts or theories:

1. **Intelligible.** Teachers should do what they can to increase the intelligibility of the new theory. Students must be able to grasp how the new conception works.

To increase intelligibility, teachers can use methods such as:

Analogies

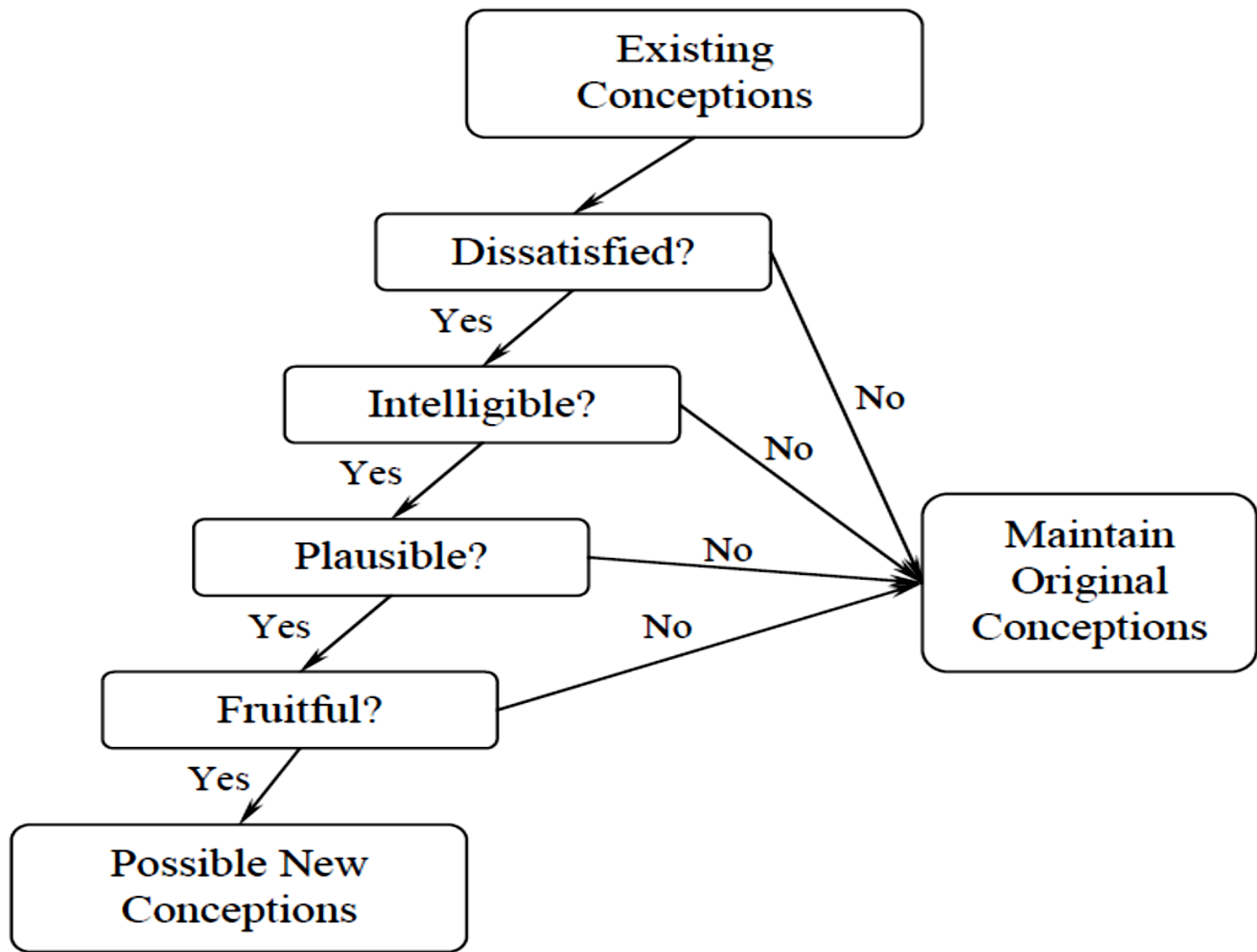
Models (both pictorial conceptual and physical)

Direct exposition (Discovery Learning)

2. Plausible. Students must see how the new conception (theory) is consistent with other knowledge and a good explanation of the data.

3. Fruitful. Students must be able to extend the new conception to new areas of inquiry.

Illustrate the application of the new concept/theory to a range of problems. These problems can include familiar ones and new ones.

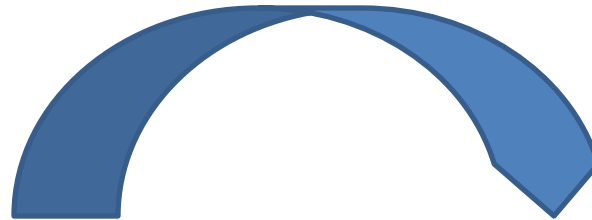


Strategies/Teaching approaches to promote Conceptual Change

Bridging Analogies

One of the best ways that teachers can correct misconceptions is by a strategy called "using bridging analogies."

Intermediate similar
or analogous examples

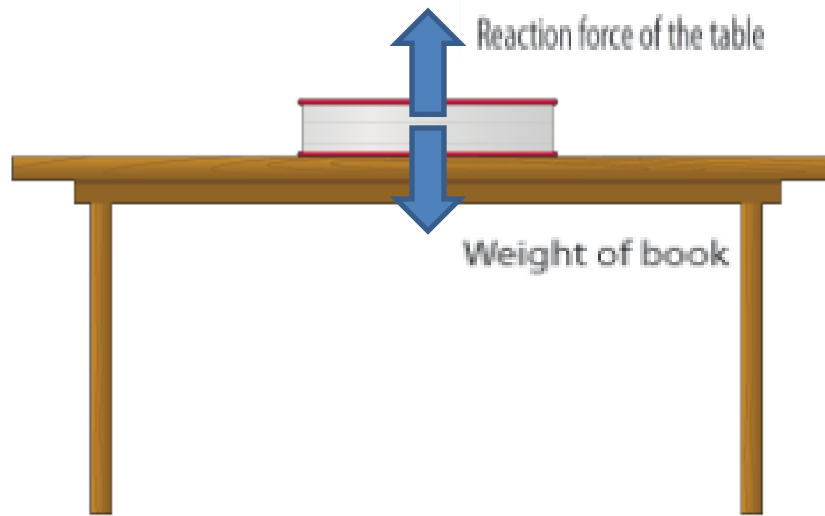


Anchoring Conceptions
Students' correct beliefs

The new concept

)

Bridging analogies: An example



Misconception:

Static objects are rigid barriers that cannot exert forces

Anchoring Conception:

A spring pushes up on one's hand when the hand is pushing down on the spring

1. Anchor example: hand on spring.



2. Bridging example 1: book (and hand) on a spring.



3. Bridging example 2: books on a flexible board.



4. Bridging example 3: hand on a flexible board.

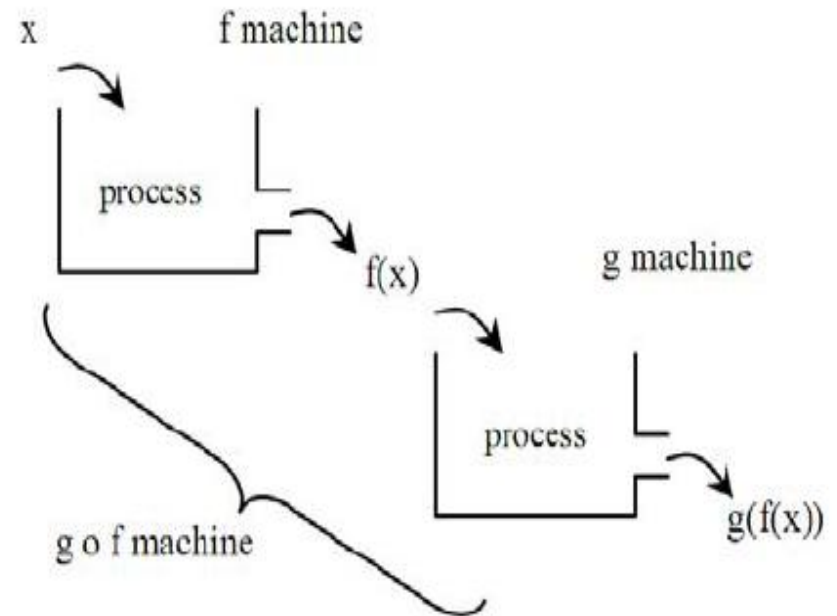


5. Target example: book on table



The “Washing machine” analogy

In the case of “washing machine” analogy the functions $f : A \rightarrow B$ and $g : B \rightarrow C$ can be composed by first applying f (washing process in washing machine) to an argument x (dirty clothes) and then applying g (drying process in a dryer) to the result



Diverse Instruction

Diverse instruction simultaneously challenges at least two erroneous beliefs that underlie a misconception.

Conceptual change is more likely if students are presented with a few examples that challenge multiple assumptions, rather than with a larger number of examples that challenge just one assumption.

Diverse Instruction: An Example

- Children's erroneous beliefs about the earth can be linked to two more general misconceptions:
 1. The belief that the earth appears flat to an observer on the ground.
 2. Misunderstanding of gravity and misconception on the influence of gravity on objects located on different parts of the earth's surface.

The shape

Thales of Miletus (624 – 546 BC)

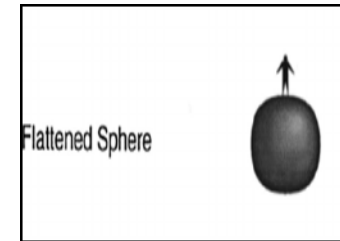
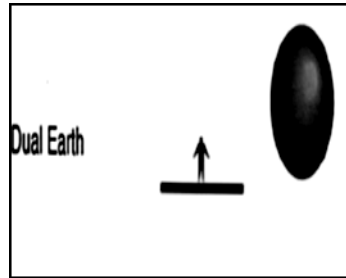
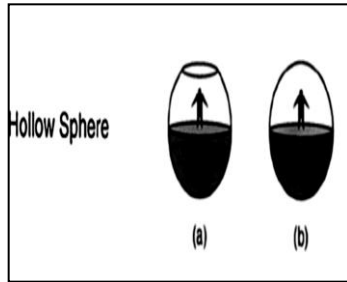


When a ship is at the horizon, its lower part is invisible due to Earth's curvature. This was one of the first arguments favouring a round-Earth model.

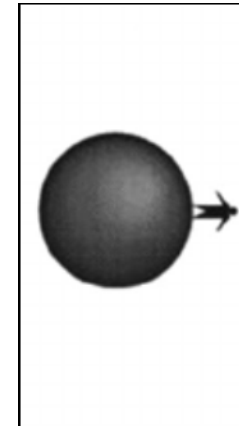
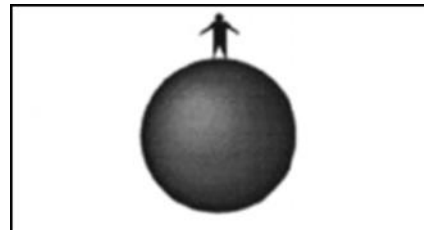
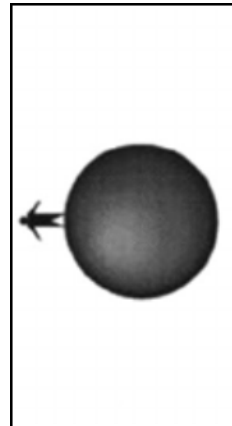
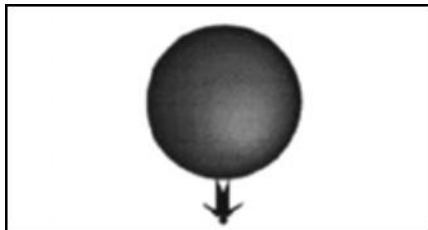


Photos from satellites

Vosniadou & Brewer (1992)



- Children receiving instruction about two core beliefs are more likely to show an increased rate of acceptance of a spherical earth model at post-test time.



Cognitive Conflict

Cognitive conflict or 'disequilibrium' arises when students encounter experiences that they are not able to assimilate or that are incompatible with their current conceptions (misconceptions).

Ways to generate cognitive conflict in the mind of the student:

1. Present students with anomalous data (data that do not accord with their misconception).

This strategy is thought to be a major means of eliciting cognitive conflict and getting students to change or abandon their current erroneous theories and adopt new ones.

Example:

Many students could not differentiate the terms heat and temperature and they use these terms interchangeably.

Anomalous Data



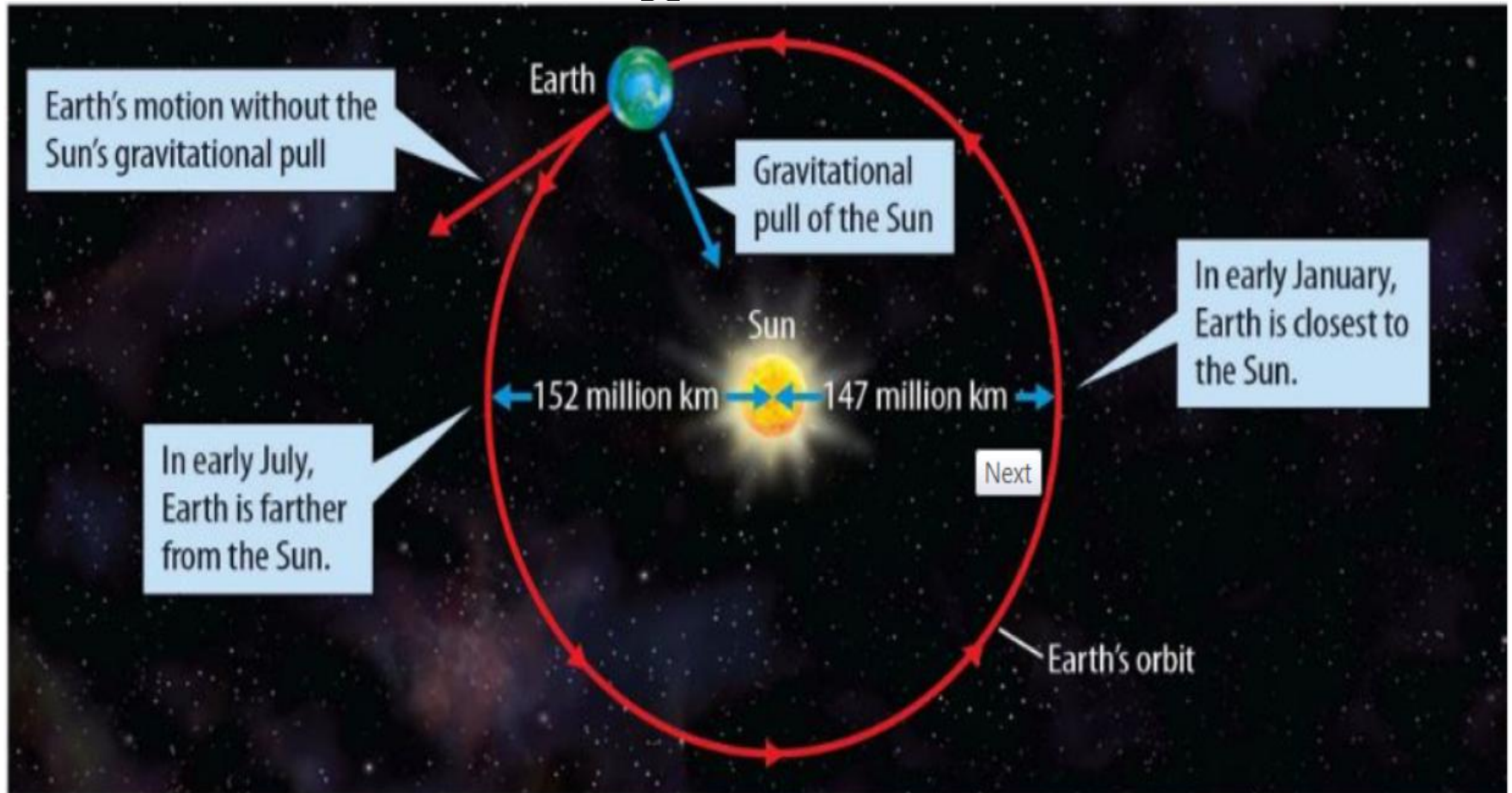
2. Present students with 'refutational' texts (texts wherein a misconception is explicitly refuted by presenting contrasting information) and/or present students with text that presents the new theory or concept.

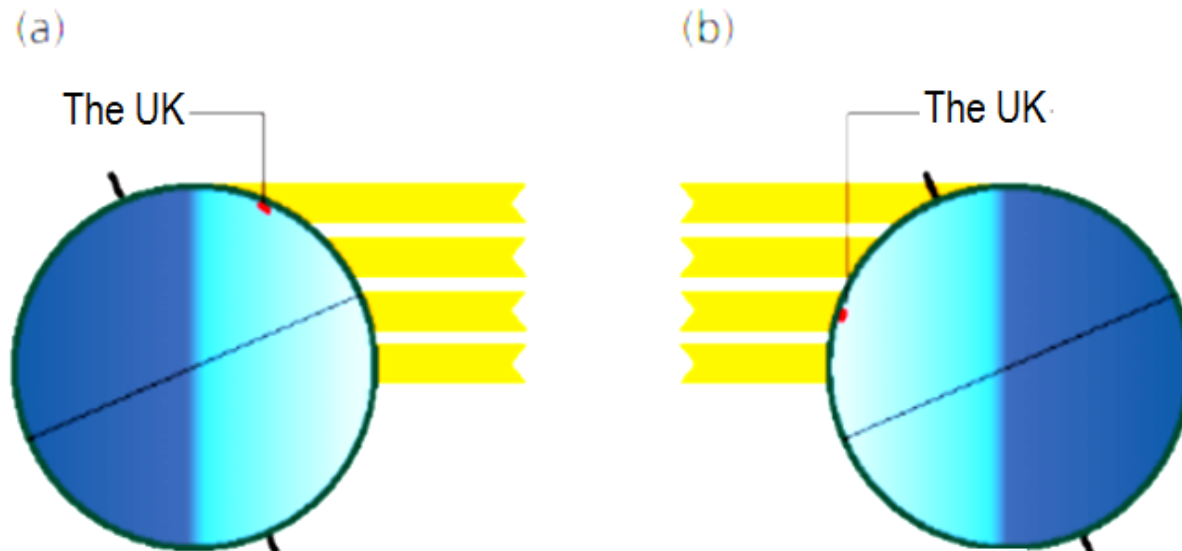
Example:

A common misconception on mechanics is that moving objects stop on their own (known as force being used up misconception)

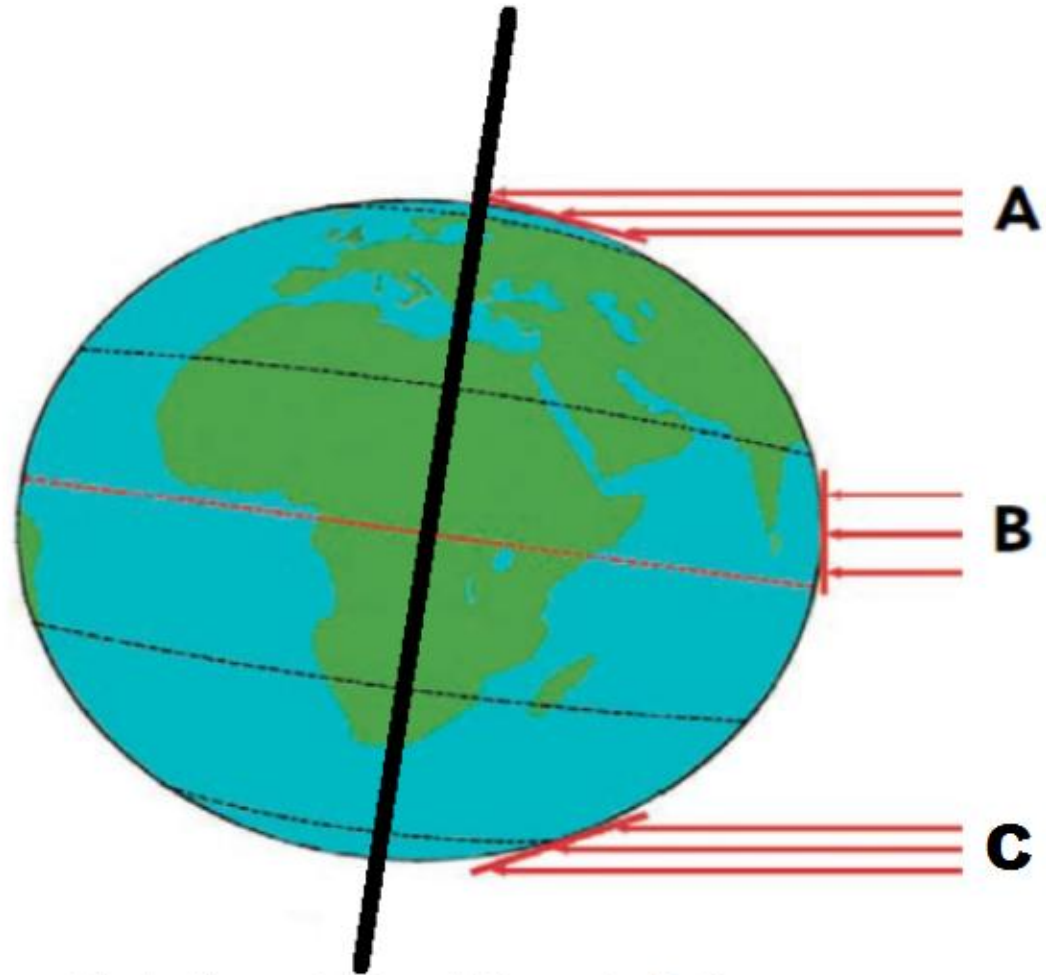
Despite the fact that many people think that a rolling ball will slow or stop on its own, this will not happen... Moving objects will keep moving at a constant rate unless they are slowed or stopped, or their direction is changed because of an outside force such as friction.

Addressing anticipated learning difficulties through conflict situations

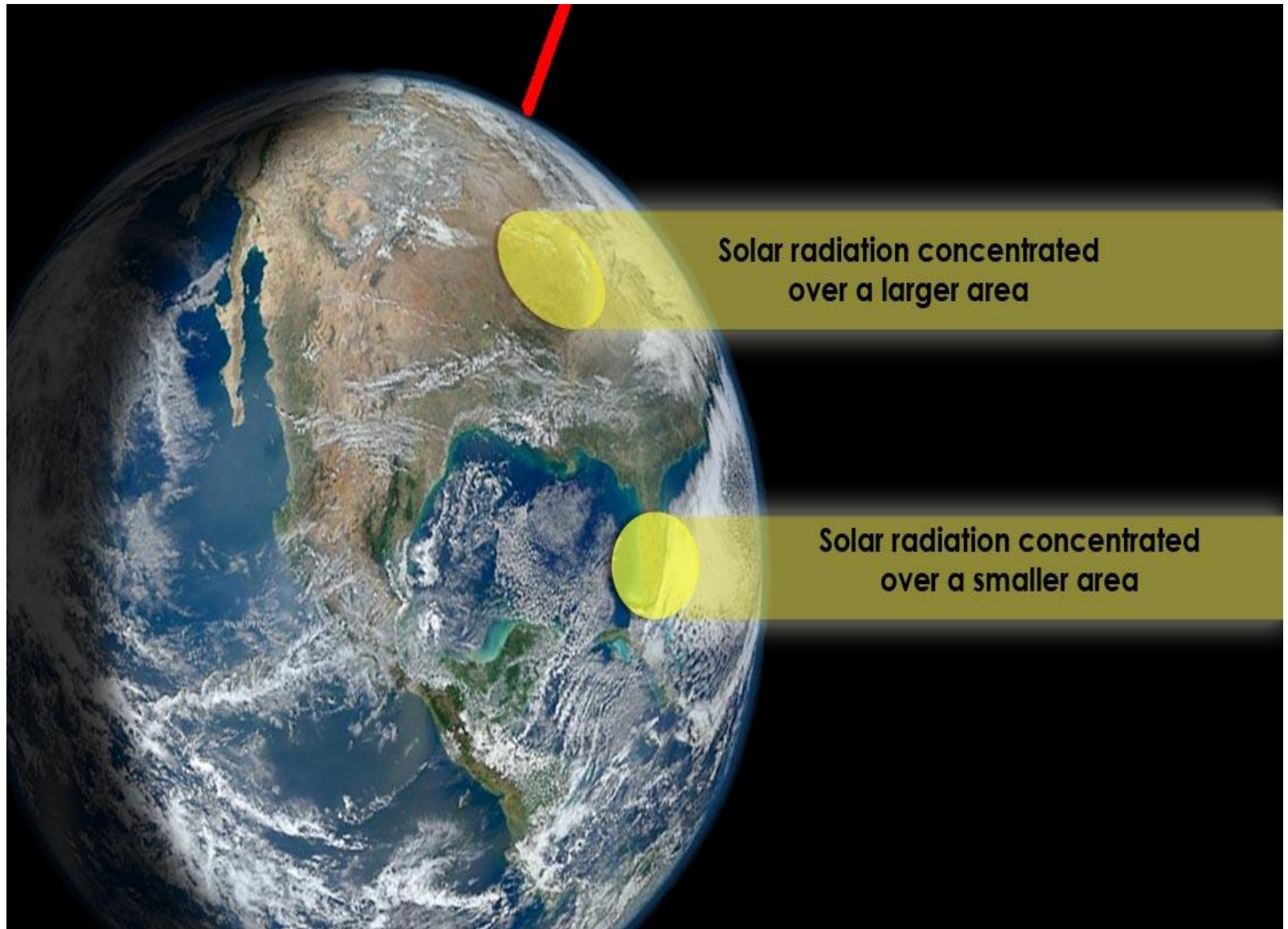




The Earth is shown (a) with its axis tilted 23.5° away from the Sun (winter), (b) with its axis tilted 23.5° towards the Sun (summer in Ireland). Ireland is represented with a red dot.



The incidence of solar radiation on the Earth.

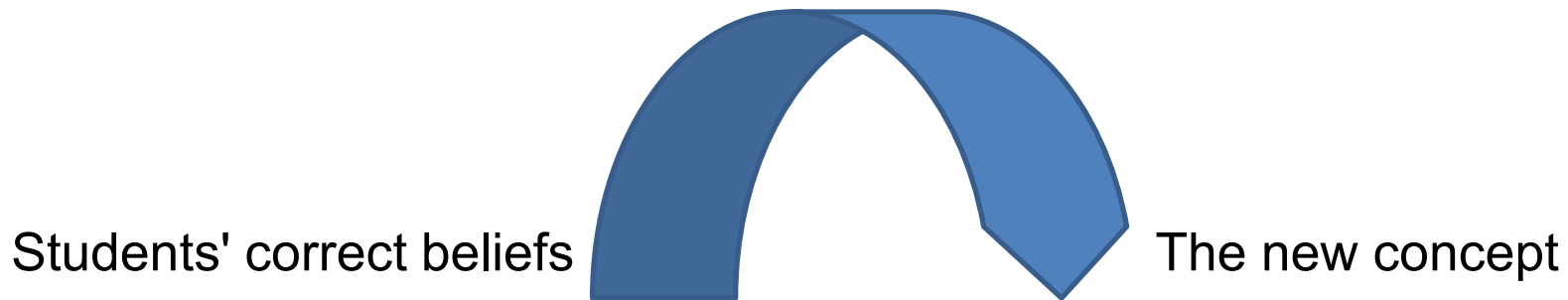


Conceptual Change

One of the best ways that teachers can correct misconceptions is by a strategy called **bridging analogies**.

*Intermediate similar
or analogous examples/experiences*

Anchoring Conceptions



Anchoring Conception

The intensity of light from a torch or a bulb that shines on a surface depends on the angle at which the beam of light strikes the surface

- <https://www.e-education.psu.edu/eme810/node/474>

‘Science Talk’ elicit
conceptual change
and helps students to
formulate ideas.

Summary

1. Students come to science class with ideas
2. Students' ideas are often different from scientists
3. Students' preconceptions are strongly held
4. Traditional instruction (rote learning) will not lead to substantial conceptual change, and
5. Effective instructional strategies enable teachers to teach for conceptual change and understanding

What *ARE* the misconceptions?

- Think about the topic you are currently teaching or have observed.
- What are the misconceptions surrounding this topic?
- How are you going to assess if your students have these misconceptions?
- Why do you need to?



Key references

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